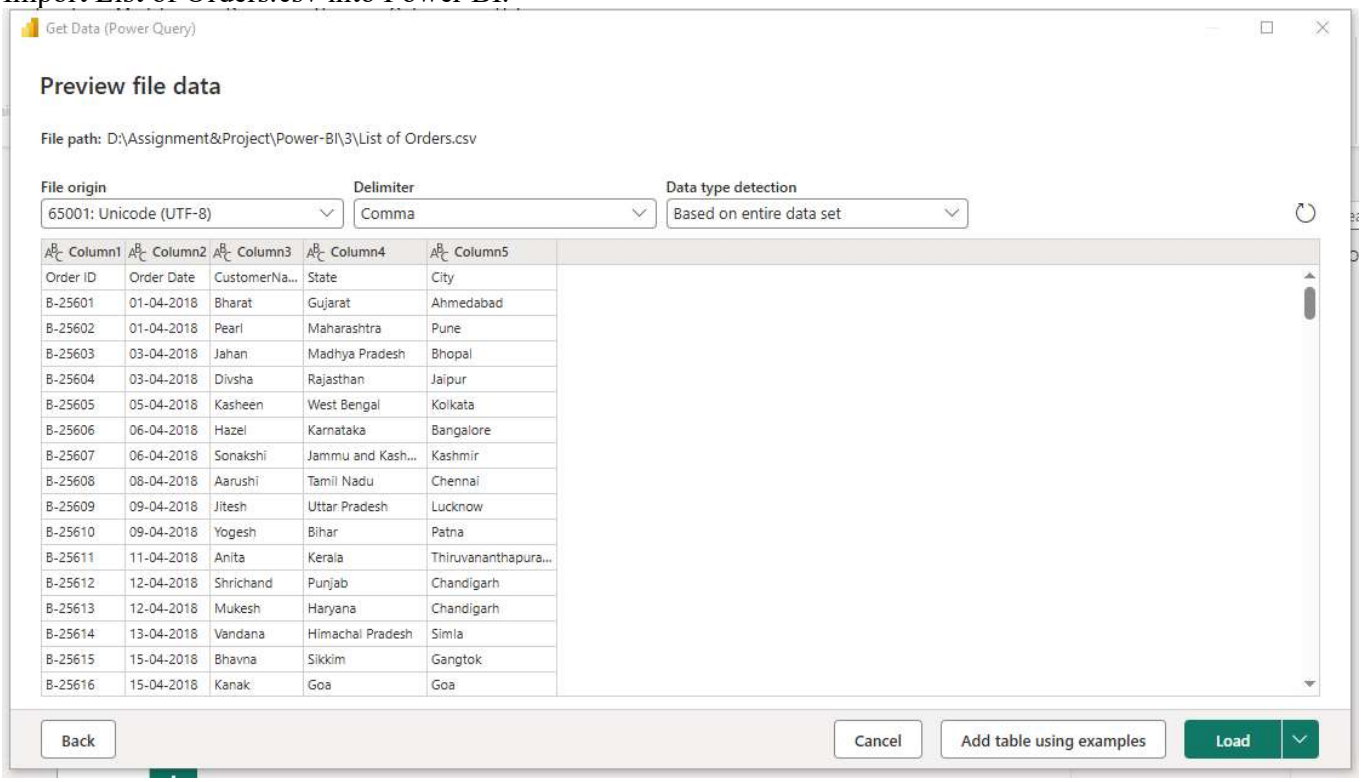
Data Transformation & Data Modelling

Data Import & Transformation

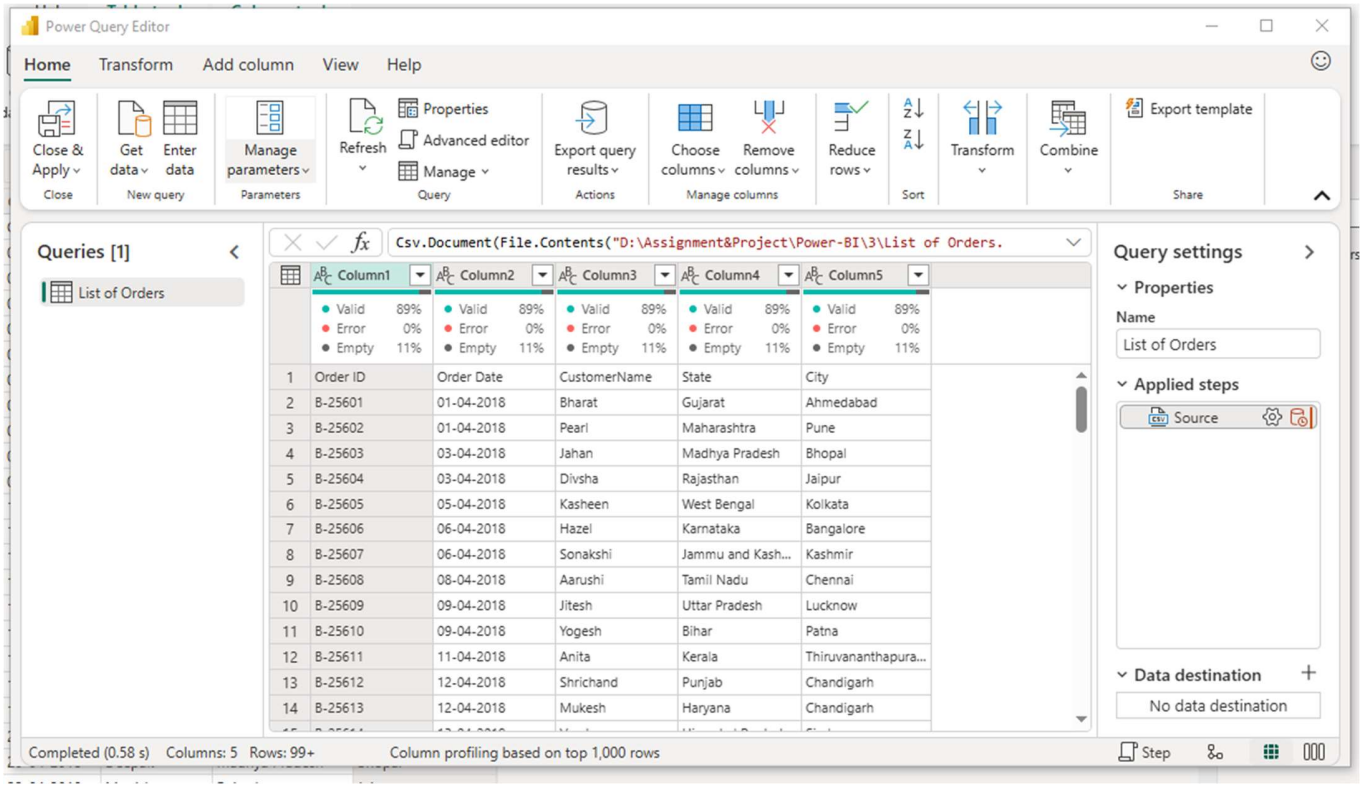
1. Data Import:

Import List of Orders.csv into Power BI.

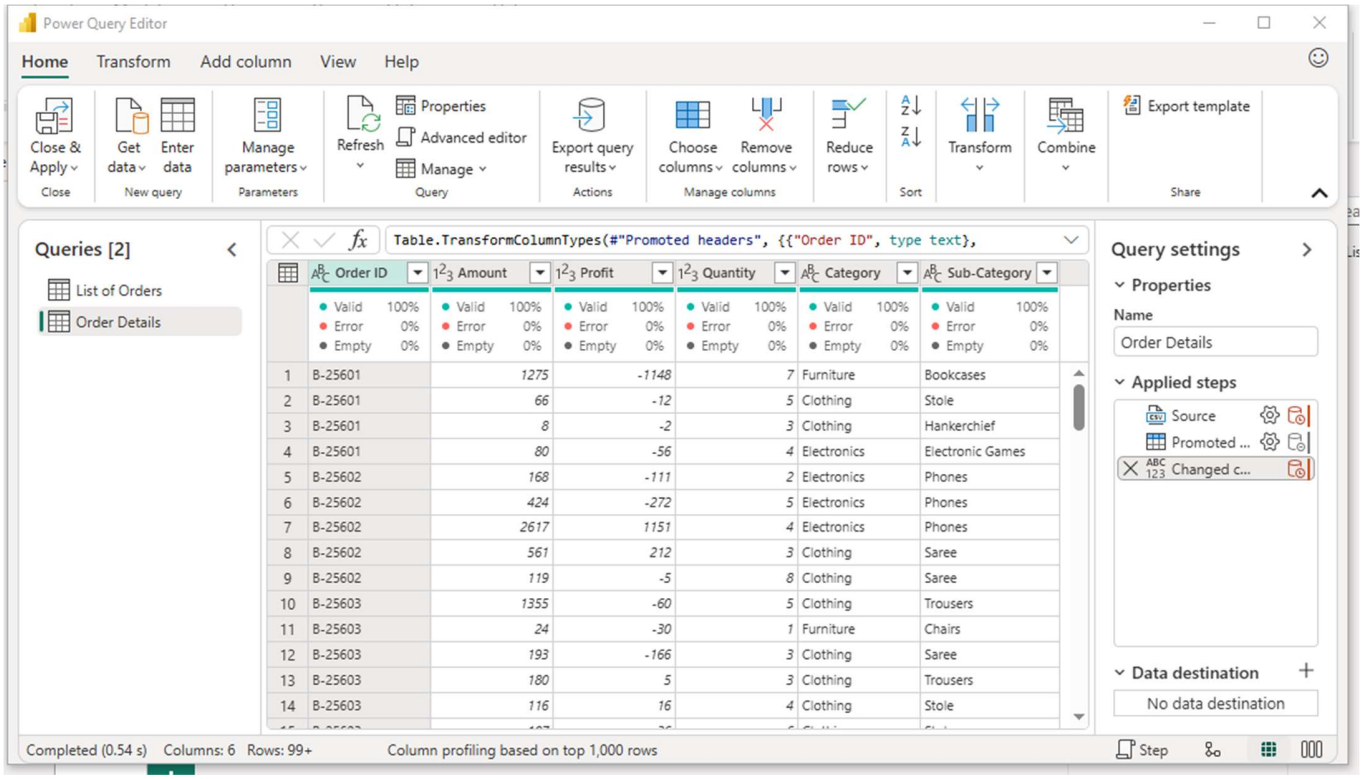


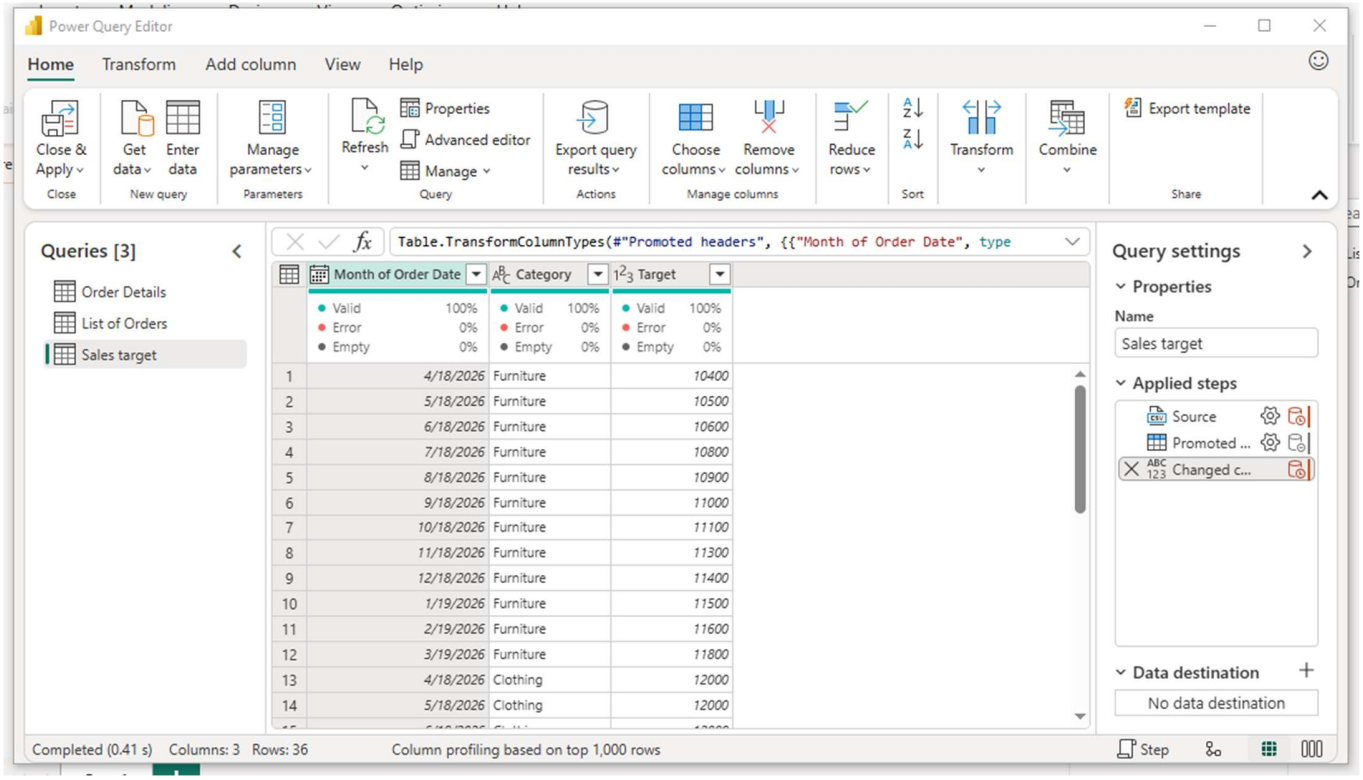
Column1	Column2	Column3	Column4	Column5
Order ID	Order Date	CustomerNa...	State	City
B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
B-25602	01-04-2018	Pearl	Maharashtra	Pune
B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
B-25606	06-04-2018	Hazel	Karnataka	Bangalore
B-25607	06-04-2018	Sonakshi	Jammu and Kash...	Kashmir
B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
B-25610	09-04-2018	Yogesh	Bihar	Patna
B-25611	11-04-2018	Anita	Kerala	Thiruvananthapura...
B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
B-25613	12-04-2018	Mukesh	Haryana	Chandigarh
B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
B-25615	15-04-2018	Bhavna	Sikkim	Gangtok
B-25616	15-04-2018	Kanak	Goa	Goa

Open the list of orders in **the Power Query Editor** using the **Transform Data** option.



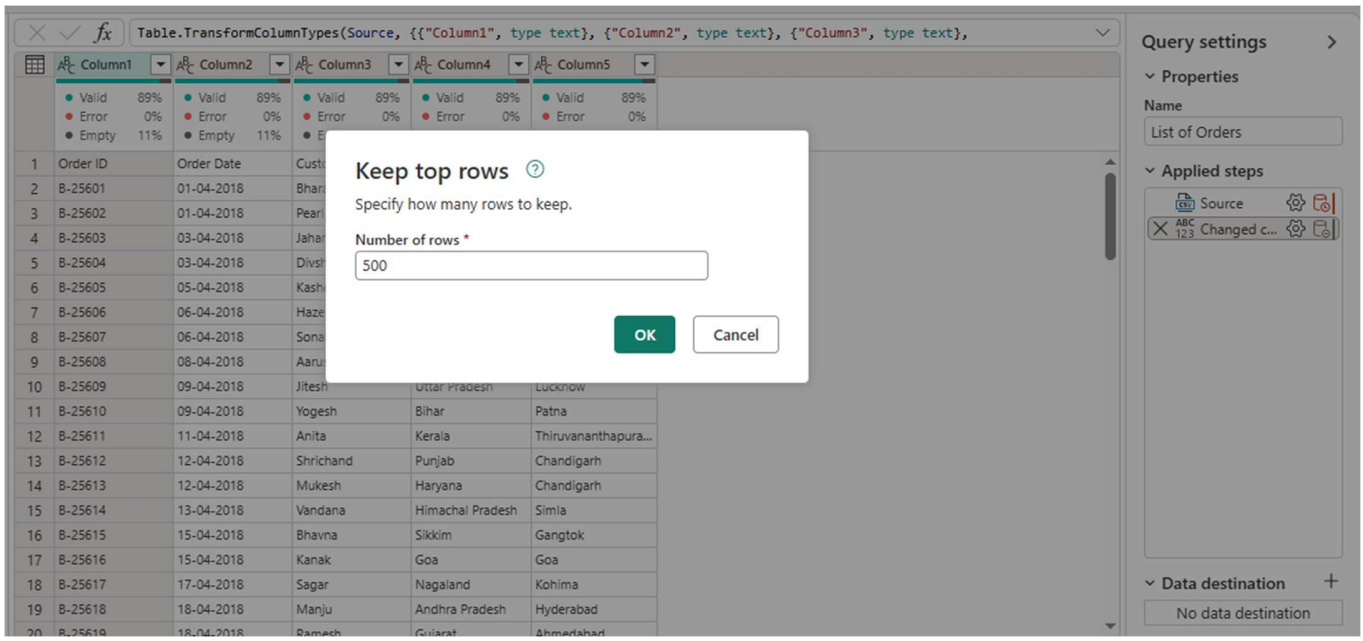
Import Order Details.csv and Sales Target.csv into Power Query Editor.



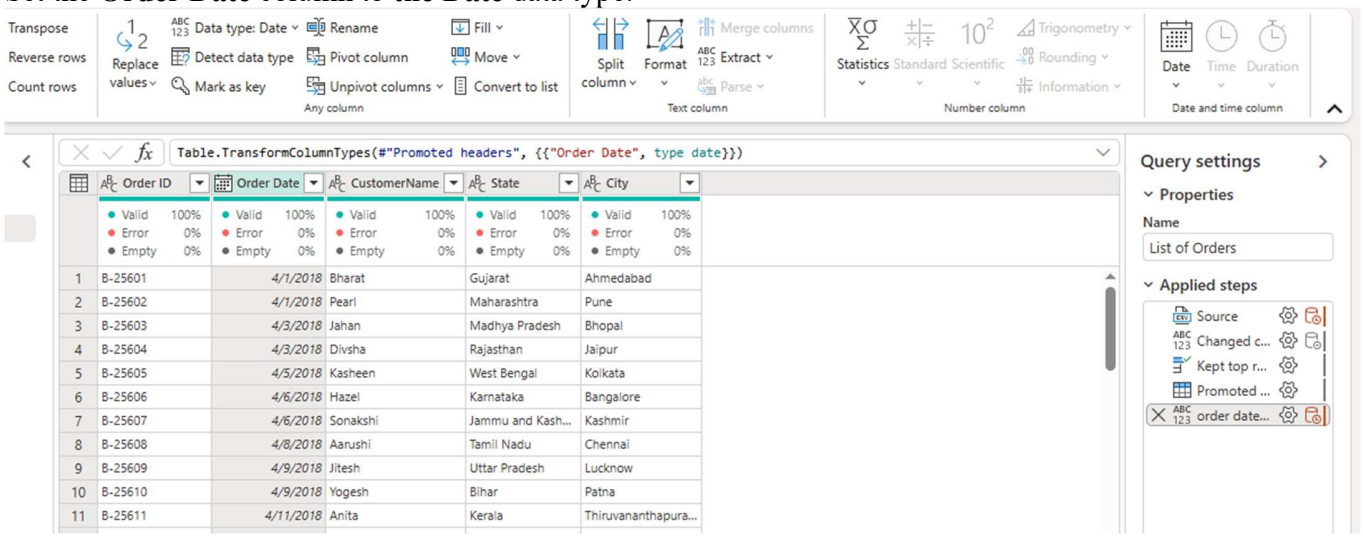


2. Row Limiting & Data Types:

Restrict the List of Orders table to the **first 500 rows**.

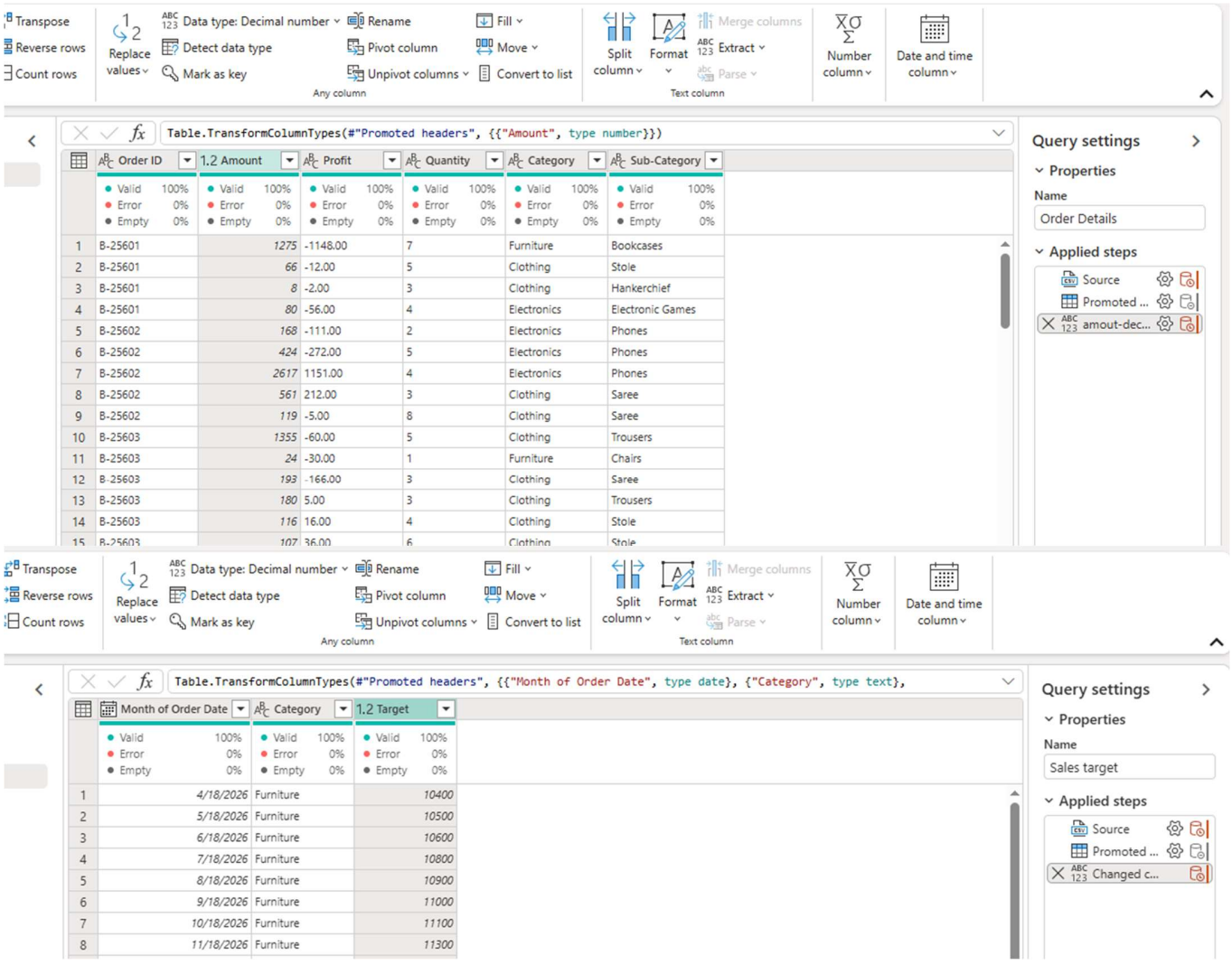


Set the **Order Date** column to **the Date** data type.



	Order ID	Order Date	CustomerName	State	City
1	B-25601	4/1/2018	Bharat	Gujarat	Ahmedabad
2	B-25602	4/1/2018	Pearl	Maharashtra	Pune
3	B-25603	4/3/2018	Jahan	Madhya Pradesh	Bhopal
4	B-25604	4/3/2018	Divsha	Rajasthan	Jaipur
5	B-25605	4/5/2018	Kasheen	West Bengal	Kolkata
6	B-25606	4/6/2018	Hazel	Karnataka	Bangalore
7	B-25607	4/6/2018	Sonakshi	Jammu and Kash...	Kashmir
8	B-25608	4/8/2018	Aarushi	Tamil Nadu	Chennai
9	B-25609	4/9/2018	Jitesh	Uttar Pradesh	Lucknow
10	B-25610	4/9/2018	Yogesh	Bihar	Patna
11	B-25611	4/11/2018	Anita	Kerala	Thiruvananthapura...

Change the **Amount** and **Target** columns to **fixed decimal numbers.**

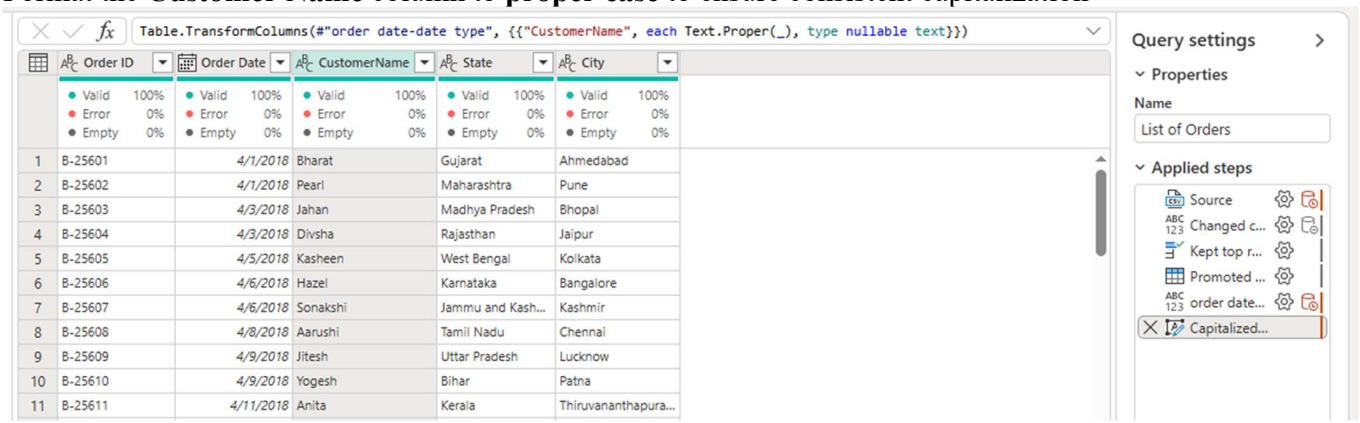


	Order ID	1.2 Amount	Profit	Quantity	Category	Sub-Category
1	B-25601	1275	-1148.00	7	Furniture	Bookcases
2	B-25601	66	-12.00	5	Clothing	Stole
3	B-25601	8	-2.00	3	Clothing	Hankerchief
4	B-25601	80	-56.00	4	Electronics	Electronic Games
5	B-25602	168	-111.00	2	Electronics	Phones
6	B-25602	424	-272.00	5	Electronics	Phones
7	B-25602	2617	1151.00	4	Electronics	Phones
8	B-25602	561	212.00	3	Clothing	Saree
9	B-25602	119	-5.00	8	Clothing	Saree
10	B-25603	1355	-60.00	5	Clothing	Trousers
11	B-25603	24	-30.00	1	Furniture	Chairs
12	B-25603	193	-166.00	3	Clothing	Saree
13	B-25603	180	5.00	3	Clothing	Trousers
14	B-25603	116	16.00	4	Clothing	Stole
15	B-25603	107	36.00	6	Clothing	Stole

	Month of Order Date	Category	1.2 Target
1	4/18/2026	Furniture	10400
2	5/18/2026	Furniture	10500
3	6/18/2026	Furniture	10600
4	7/18/2026	Furniture	10800
5	8/18/2026	Furniture	10900
6	9/18/2026	Furniture	11000
7	10/18/2026	Furniture	11100
8	11/18/2026	Furniture	11300

3.Text Formatting:

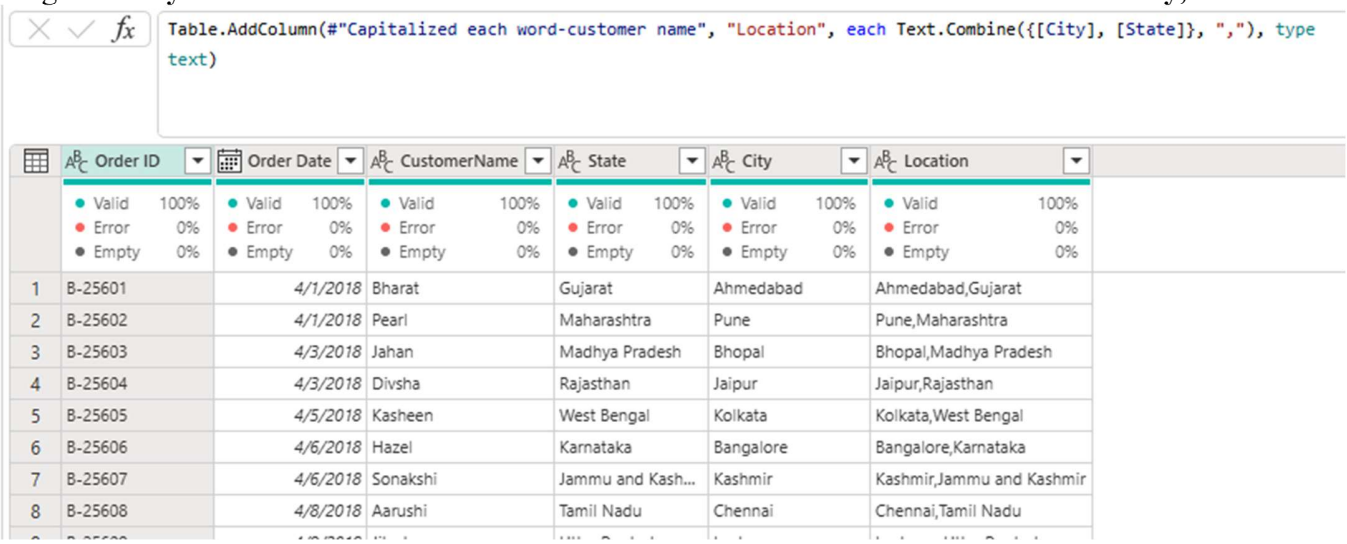
Format the **Customer Name** column to **proper case** to ensure consistent capitalization



	Order ID	Order Date	CustomerName	State	City
1	B-25601	4/1/2018	Bharat	Gujarat	Ahmedabad
2	B-25602	4/1/2018	Pearl	Maharashtra	Pune
3	B-25603	4/3/2018	Jahan	Madhya Pradesh	Bhopal
4	B-25604	4/3/2018	Divsha	Rajasthan	Jaipur
5	B-25605	4/5/2018	Kasheen	West Bengal	Kolkata
6	B-25606	4/6/2018	Hazel	Karnataka	Bangalore
7	B-25607	4/6/2018	Sonakshi	Jammu and Kash...	Kashmir
8	B-25608	4/8/2018	Aarushi	Tamil Nadu	Chennai
9	B-25609	4/9/2018	Jitesh	Uttar Pradesh	Lucknow
10	B-25610	4/9/2018	Yogesh	Bihar	Patna
11	B-25611	4/11/2018	Anita	Kerala	Thiruvananthapura...

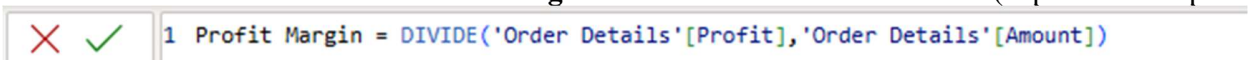
4. Column Creation:

Merge the **City** and **State** columns to create a new column named **Location** in the format: **City, State**



	Order ID	Order Date	CustomerName	State	City	Location
1	B-25601	4/1/2018	Bharat	Gujarat	Ahmedabad	Ahmedabad,Gujarat
2	B-25602	4/1/2018	Pearl	Maharashtra	Pune	Pune,Maharashtra
3	B-25603	4/3/2018	Jahan	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh
4	B-25604	4/3/2018	Divsha	Rajasthan	Jaipur	Jaipur,Rajasthan
5	B-25605	4/5/2018	Kasheen	West Bengal	Kolkata	Kolkata,West Bengal
6	B-25606	4/6/2018	Hazel	Karnataka	Bangalore	Bangalore,Karnataka
7	B-25607	4/6/2018	Sonakshi	Jammu and Kash...	Kashmir	Kashmir,Jammu and Kashmir
8	B-25608	4/8/2018	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu

Create a custom column named **Profit Margin** calculated as: **Profit / Amount** (expressed as a percentage)



Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin
B-25602	561	212.00	3	Clothing	Saree	37.79%
B-25602	119	-5.00	8	Clothing	Saree	-4.20%
B-25603	193	-166.00	3	Clothing	Saree	-86.01%
B-25604	157	5.00	9	Clothing	Saree	3.18%
B-25605	75	0.00	7	Clothing	Saree	0.00%
B-25609	25	-5.00	4	Clothing	Saree	-20.00%
B-25610	43	0.00	3	Clothing	Saree	0.00%

5. Conditional Column:

Add a conditional column named **Profit Status** based on:
■ Profit < 0 → Loss ■ Profit = 0 → Break-Even ■ Profit > 0 → Profit

```
Table.AddColumn(#"Changed column type", "Profit Status", each if [Profit] < 0 then "Low" else if [Profit] = 0 then
"Break-Even" else "Profit")
```

	Order ID	1.2 Amount	1.2 Profit	Quantity	Category	Sub-Category	Profit Status
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	B-25601	1275	-1148	7	Furniture	Bookcases	Low
2	B-25601	66	-12	5	Clothing	Stole	Low
3	B-25601	8	-2	3	Clothing	Hankerchief	Low
4	B-25601	80	-56	4	Electronics	Electronic Games	Low
5	B-25602	168	-111	2	Electronics	Phones	Low
6	B-25602	424	-272	5	Electronics	Phones	Low
7	B-25602	2617	1151	4	Electronics	Phones	Profit

6. Merging Data (Joins)

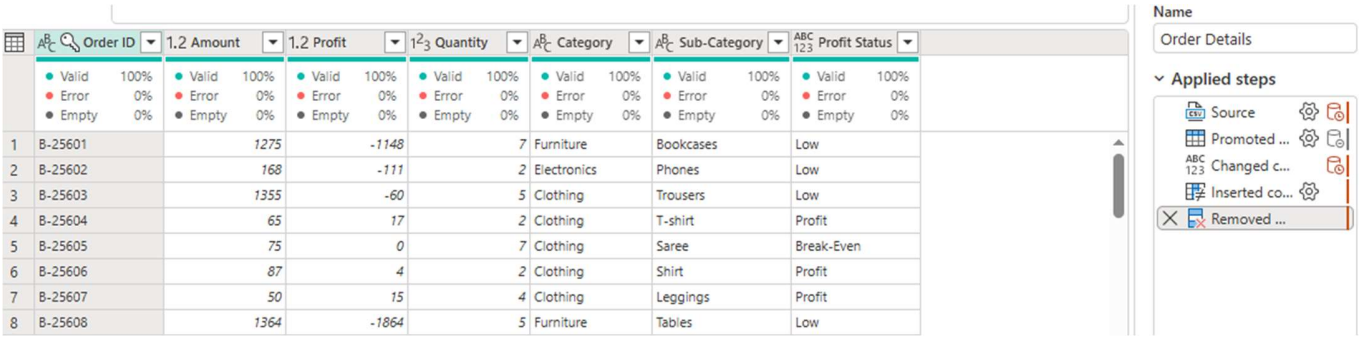
Merge the List of Orders and Order Details tables using **Order ID**
Name the merged output table as **Orders Data**.

Order Details
List of Orders
Sales target
Orders Data

	Order ID	Order Date	CustomerName	State	City	Location	Order Details
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	N/A for this column type
1	B-25601	4/1/2018	Bharat	Gujarat	Ahmedabad	Ahmedabad,Gujarat	[Table]
2	B-25602	4/1/2018	Pearl	Maharashtra	Pune	Pune,Maharashtra	[Table]
3	B-25603	4/3/2018	Jahan	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh	[Table]
4	B-25604	4/3/2018	Divsha	Rajasthan	Jaipur	Jaipur,Rajasthan	[Table]
5	B-25605	4/5/2018	Kasheen	West Bengal	Kolkata	Kolkata,West Bengal	[Table]
6	B-25606	4/6/2018	Hazel	Karnataka	Bangalore	Bangalore,Karnataka	[Table]
7	B-25607	4/6/2018	Sonakshi	Jammu and Kash...	Kashmir	Kashmir,Jammu and Kashmir	[Table]
8	B-25608	4/8/2018	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	[Table]
9	B-25609	4/9/2018	Jitesh	Uttar Pradesh	Lucknow	Lucknow,Uttar Pradesh	[Table]
10	B-25610	4/9/2018	Yogesh	Bihar	Patna	Patna,Bihar	[Table]
11	B-25611	4/11/2018	Anita	Kerala	Thiruvananthapura...	Thiruvananthapuram,Kerala	[Table]
12	B-25612	4/12/2018	Shrichand	Punjab	Chandigarh	Chandigarh,Punjab	[Table]

7. Handling Missing Data & Duplicate Data

- Identify missing values across all tables.
- Define and apply appropriate strategies such as:
  - Replacing missing values
  - Removing records
  - Retaining nulls with justification
- Identify duplicate rows and define a suitable handling strategy based on business logic.



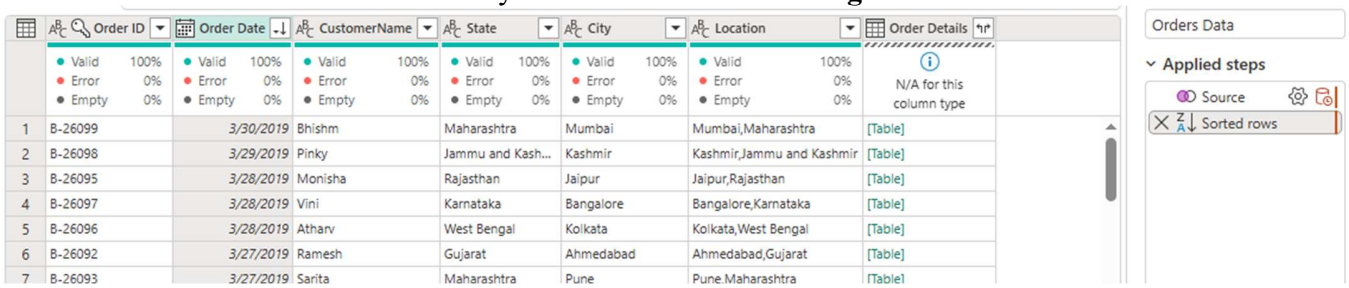
	Order ID	1.2 Amount	1.2 Profit	1²3 Quantity	Category	Sub-Category	Profit Status
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	B-25601	1275	-1148	7	Furniture	Bookcases	Low
2	B-25602	168	-111	2	Electronics	Phones	Low
3	B-25603	1355	-60	5	Clothing	Trousers	Low
4	B-25604	65	17	2	Clothing	T-shirt	Profit
5	B-25605	75	0	7	Clothing	Saree	Break-Even
6	B-25606	87	4	2	Clothing	Shirt	Profit
7	B-25607	50	15	4	Clothing	Leggings	Profit
8	B-25608	1364	-1864	5	Furniture	Tables	Low

Name: Order Details

Applied steps: Source, Promoted ..., Changed c..., Inserted co..., Removed ...

8. Sorting and Filtering Data

Sort records in the **Orders Data** table by **Order Date** in **descending order**.



	Order ID	Order Date	CustomerName	State	City	Location	Order Details
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	N/A for this column type
1	B-26099	3/30/2019	Bhishm	Maharashtra	Mumbai	Mumbai,Maharashtra	[Table]
2	B-26098	3/29/2019	Pinky	Jammu and Kash...	Kashmir	Kashmir,Jammu and Kashmir	[Table]
3	B-26095	3/28/2019	Monisha	Rajasthan	Jaipur	Jaipur,Rajasthan	[Table]
4	B-26097	3/28/2019	Vini	Karnataka	Bangalore	Bangalore,Karnataka	[Table]
5	B-26096	3/28/2019	Atharv	West Bengal	Kolkata	Kolkata,West Bengal	[Table]
6	B-26092	3/27/2019	Ramesh	Gujarat	Ahmedabad	Ahmedabad,Gujarat	[Table]
7	B-26093	3/27/2019	Sarita	Maharashtra	Pune	Pune,Maharashtra	[Table]

Orders Data

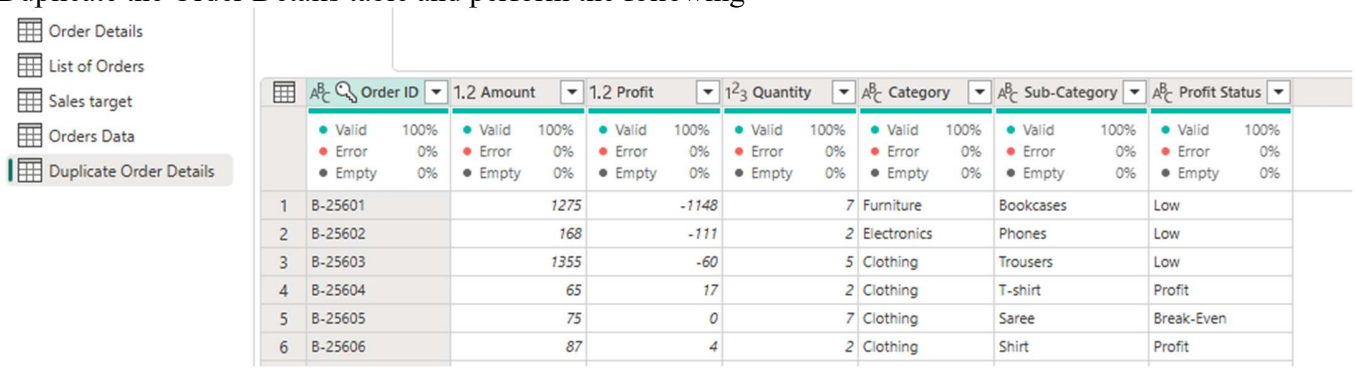
Applied steps: Source, Sorted rows

Apply filters to analyze orders from a specific state (e.g., **Tamil Nadu**).

	Order ID	Order Date	CustomerName	State	City	Location	Order Details
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	N/A for this column type
1	B-26081	3/22/2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	[Table]
2	B-26018	2/14/2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	[Table]
3	B-26008	2/9/2019	Kalyani	Tamil Nadu	Chennai	Chennai,Tamil Nadu	[Table]
4	B-25860	11/15/2018	Akshay	Tamil Nadu	Chennai	Chennai,Tamil Nadu	[Table]
5	B-25788	9/21/2018	Dinesh	Tamil Nadu	Chennai	Chennai,Tamil Nadu	[Table]
6	B-25716	7/11/2018	Surabhi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	[Table]
7	B-25698	6/23/2018	Amisha	Tamil Nadu	Chennai	Chennai,Tamil Nadu	[Table]
8	B-25608	4/8/2018	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	[Table]

9. Grouping and Aggregating Data

Duplicate the Order Details table and perform the following



	Order ID	1.2 Amount	1.2 Profit	Quantity	Category	Sub-Category	Profit Status
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	B-25601	1275	-1148	7	Furniture	Bookcases	Low
2	B-25602	168	-111	2	Electronics	Phones	Low
3	B-25603	1355	-60	5	Clothing	Trousers	Low
4	B-25604	65	17	2	Clothing	T-shirt	Profit
5	B-25605	75	0	7	Clothing	Saree	Break-Even
6	B-25606	87	4	2	Clothing	Shirt	Profit

Count of Order ID

```
List.NonNullCount(#"Changed column type 1"[Order ID])
```

500

Average Profit by Category

```
1 Average_ProfitbyCategory = SUMMARIZE('Duplicate Order details','Duplicate Order details'[Category],"Avg Profit",AVERAGE('Duplicate Order details'[Amount]))
```

Category	Avg Profit
Furniture	452.11
Electronics	525.55
Clothing	143.38

Total Amount by Sub-Category

```
1 Total Amountby Sub Cate = SUMMARIZE('Duplicate Order details','Duplicate Order details'[Sub-Category],"Total Amount",SUM('Duplicate Order details'[Amount]))
```

Sub-Category	Total Amount
Bookcases	25800
Phones	11691
Trousers	7109
T-shirt	3205
Saree	17831
Shirt	2535
Leggings	851
Tables	3967
Electronic Games	11942
Printers	16822
Chairs	5724
Hankerchief	4362
Stole	7583
Skirt	822
Furnishings	4295
Accessories	9472
Kurti	1154

Duplicate the Sales Target table and aggregate the total Target by **Month of order date**

```
1 Targetby_MoO = SUMMARIZECOLUMNS('Duplicate Sales target'[Month of Order Date].[Month],"Target by Month",sum('Duplicate Sales target'[Target]))
```

Month	Target by Month
January	43500
February	43600
March	43800
April	31400
May	31500
June	31600
July	33800
August	33900
September	34000
October	36100
November	36300
December	36400

10. Data Modelling:

Establish a relationship between:

List of Orders and Order Details using **Order ID,** Order Details and Sales Target using **Category**

